

Measures of Dispersion

The following data provide the runs scored by two batsmen in the last 10 matches.

Batsman A: 25, 20, 45, 93, 8, 14, 32, 87, 72, 4

Batsman B: 33, 50, 47, 38, 45, 40, 36, 48, 37, 26

$$\text{Mean of Batsman A} = \frac{25 + 20 + 45 + 93 + 8 + 14 + 32 + 87 + 72 + 4}{10} = 40$$

$$\text{Mean of Batsman B} = \frac{33 + 50 + 47 + 38 + 45 + 40 + 36 + 48 + 37 + 26}{10} = 40$$

The mean of both data are same (40), but they differ significantly.

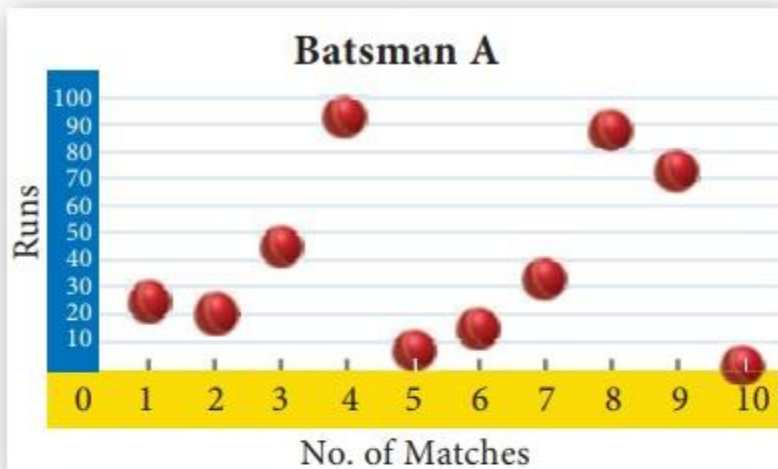


Fig. 8.1(a)

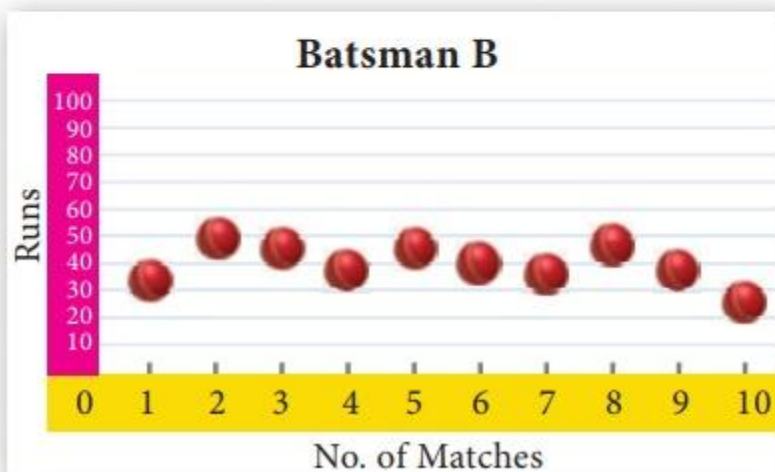


Fig. 8.1(b)

From the above diagrams, we see that runs of batsman *B* are grouped around the mean. But the runs of batsman *A* are scattered from 0 to 100, though they both have same mean.

Thus, some additional statistical information may be required to determine how the values are spread in data. For this, we shall discuss **Measures of Dispersion**.

Dispersion is a measure which gives an idea about the scatteredness of the values.

Measures of Variation (or) Dispersion of a data provide an idea of how observations spread out (or) scattered throughout the data.

Objectives of Measuring Dispersion

Measures of variations are calculated to serve the following purposes:

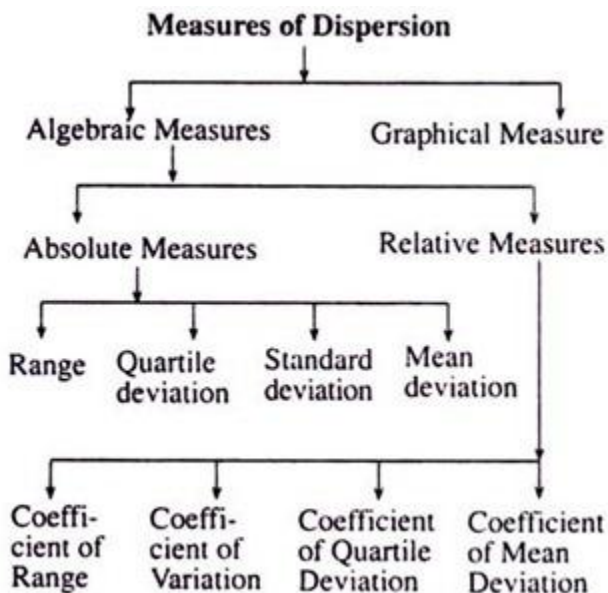
1. To judge the reliability of measures of central tendency.
2. To compare two or more series with regard to their variability.
3. To control the variability itself.
4. To facilitate the use of other statistical measure.

Characteristics of a good measure of dispersion:

An ideal measure of dispersion is expected to possess the following properties

1. It should be rigidly defined
2. It should be based on all the items.
3. It should not be unduly affected by extreme items
4. It should lend itself for algebraic manipulation.
5. It should be simple to understand and easy to calculate

Different Measures of Dispersion are



Absolute Measures of Dispersion:

These measures give us an idea about the amount of dispersion in a set of observations. They give the answers in the same units as the units of the original observations. When the observations are in kilograms, the absolute measure is also in kilograms. If we have two sets of observations, we cannot always use the absolute measures to compare their dispersion. We shall explain later as to when the absolute measures can be used for comparison of dispersion in two or more than two sets of data. The absolute measures which are commonly used are:

1. The Range
2. The Quartile Deviation
3. The Mean Deviation
4. The Standard deviation and Variance

Relative Measure of Dispersion:

These measures are calculated for the comparison of dispersion in two or more than two sets of observations. These measures are free of the units in which the original data is measured. If the original data is in dollar or kilometers, we do not use these units with relative measure of dispersion. These measures are a sort of ratio and are called coefficients. Each absolute measure of dispersion can be converted into its relative measure. Thus the relative measures of dispersion are:

1. Coefficient of Range or Coefficient of Dispersion.
2. Coefficient of Quartile Deviation or Quartile Coefficient of Dispersion.
3. Coefficient of Mean Deviation or Mean Deviation of Dispersion.
4. Coefficient of Variation

Distinguish between absolute measure and relative measure

The Absolute Measure of dispersion is basically the measure of variation from the mean such as standard deviation. On the other hand the relative measure of dispersion is basically the position of a certain variable with reference to or as compared with the other variables. Such as Coefficient of Variation. Absolute measures of Dispersion are expressed in same units in which original data is presented but these measures cannot be used to compare the variations between the two series. Relative measures are not expressed in units but it is a pure number. It is the ratios of absolute dispersion to an appropriate average such as co-efficient of Standard Deviation or Co-efficient of Mean Deviation. Absolute measure does not expressed in percentage relative measure is usually expressed in percentage.

Range

The difference between the largest value and the smallest value is called Range.

$$\text{Range } R = L - S$$

where L - Largest value; S - Smallest value

Example Find the range of the following data: 25, 67, 48, 53, 18, 39, 44.

Solution Largest value $L = 67$; Smallest value $S = 18$

$$\text{Range } R = L - S = 67 - 18 = 49$$

Merits and Demerits of Range:

Merits:

1. It is simple to understand.
2. It is easy to calculate.
3. In certain types of problems like quality control, weather forecasts, share price analysis, etc., range is most widely used.

Demerits:

1. It is very much affected by the extreme items.
2. It is based on only two extreme observations.
3. It cannot be calculated from open-end class intervals.
4. It is not suitable for mathematical treatment.
5. It is a very rarely used measure.

Coefficient of range :

$$\text{Coefficient of range} = [(L - S) / (L + S)] \times 100$$

Where L - Largest value; S - Smallest value

Example Find the coefficient of range of the following data: 25, 67, 48, 53, 18, 39, 44.

Solution Largest value $L = 67$; Smallest value $S = 18$

$$\text{Range } R = L - S = 67 - 18 = 49$$

$$\text{Coefficient of range} = [(L - S) / (L + S)] \times 100$$

$$\text{Coefficient of range} = [(67 - 18) / (67 + 18)] \times 100 = [49/85] \times 100$$

$$= 0.576 \times 100 = 57.6\%$$

Example Find the range of the following distribution.

Age (in years)	16-18	18-20	20-22	22-24	24-26	26-28
Number of students	0	4	6	8	2	2

Solution Here Largest value $L = 28$

Smallest value $S = 18$

Range $R = L - S$

$R = 28 - 18 = 10$ Years

Example The range of a set of data is 13.67 and the largest value is 70.08. Find the smallest value.

Solution

Range $R = 13.67$

Largest value $L = 70.08$

Range $R = L - S$

$13.67 = 70.08 - S$

$S = 70.08 - 13.67 = 56.41$

Therefore, the smallest value is 56.41.

Quartile Deviation:

It is based on the lower quartile Q_1 and the upper quartile Q_3 . The difference $Q_3 - Q_1$ is called the inter quartile range. The difference $Q_3 - Q_1$ divided by 2 is called semi-inter-quartile range

or the quartile deviation.

$$\text{Quartile Deviation} = \frac{\text{Third Quartile} - \text{First Quartile}}{2}$$

$$\text{Quartile Deviation (Q.D)} = \frac{Q_3 - Q_1}{2}$$

The quartile deviation is a slightly better measure of absolute dispersion than the range. But it ignores the observation on the tails. If we take difference samples from a population and calculate their quartile deviations, their values are quite likely to be sufficiently different. This is called sampling fluctuation. It is not a popular measure of dispersion. The quartile deviation calculated from the sample data does not help us to draw any conclusion (inference) about the quartile deviation in the population.

Merits and Demerits of Quartile Deviation

Merits:

1. It is Simple to understand and easy to calculate
2. It is not affected by extreme values.
3. It can be calculated for data with open end classes also.

Demerits:

1. It is not based on all the items. It is based on two Positional values Q_1 and Q_3 and ignores the extreme 50% of the items
2. It is not amenable to further mathematical treatment.
3. It is affected by sampling fluctuations.

Coefficient of Quartile Deviation:

A relative measure of dispersion based on the quartile deviation is called the coefficient of quartile deviation. It is defined as

$$\text{Coefficient of Quartile Deviation} = \frac{\frac{Q_3 - Q_1}{\frac{Q_3 + Q_1}{2}}}{2} \times 100 = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

It is pure number free of any units of measurement. It can be used for comparing the dispersion in two or more than two sets of data.

Example 1:

12 15 20 28 30 40 50

n = number of observations = 7

Q_1 = Size of $\left(\frac{(n+1)}{4}\right)^{th}$ value = Size of $\left(\frac{7+1}{4}\right)^{th}$ value = Size of 2nd value = 15

Hence $Q_1 = 15$

Q_3 = Size of $\left(\frac{3(n+1)}{4}\right)^{th}$ value = Size of $\left(\frac{3 \times 8}{4}\right)^{th}$ value = Size of 6th value = 40

Hence $Q_3 = 40$

$$QD = \frac{1}{2} (Q_3 - Q_1) = \frac{40 - 15}{2} = 12.5$$

$$\text{Coefficient of } QD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 15}{40 + 15} = \frac{25}{55} = 0.455$$

$$\text{Coefficient of } QD = 0.455 \times 100 = 45.5\%$$

Example 2:

Marks	Frequency	Cumulative Frequency
X	f	cf
10	4	4
20	7	11
30	15	26
40	8	34
50	7	41
60	2	43
		$N = \sum f = 43$

Table : 8.11

$$Q_1 = \text{Size of } \left(\frac{N+1}{4} \right)^{\text{th}} \text{ value} = \text{Size } 11^{\text{th}} \text{ value} = 20$$

$$Q_3 = \text{Size of } \left(\frac{3(N+1)}{4} \right)^{\text{th}} \text{ value} = \text{Size of } 33^{\text{rd}} \text{ value} = 40$$

$$QD = \frac{1}{2} (Q_3 - Q_1) = \frac{40 - 20}{2} = 10$$

$$\begin{aligned} \text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 20}{40 + 20} = \frac{20}{60} \\ &= 0.333 \end{aligned}$$

Coefficient of QD= 0.333×100=33.3 %

Example 3:

Find quartile deviation and comment on the distribution:

Mid-value (in class)	14	16	18	20	22	24	26
No of students:	12	14	50	60	32	20	10

Solution:

Calculation

Mid-value (in class)	Class	f	cf
14	13 – 15	12	12
16	15 – 17	14	26
18	17 – 19	50	76
20	19 – 21	60	136
22	21 – 23	32	168
24	23 – 25	20	188
26	25 – 27	10	198
		N = 198	

$\therefore Q_1 = N/4 \times 1 = 198/4 \times 1 = 49.5^{\text{th}}$ item, lies on 17 – 19.

$$\therefore Q_1 = L + \frac{\frac{N}{4} \times 1 - \text{pcf}}{f} \times c = 17 + \frac{49.5 - 26}{50} \times 2 = 17.94$$

$Q_3 = N/4 \times 3 = 198/4 \times 3 = 148.5^{\text{th}}$ item, lies on 21 – 23.

$$\therefore Q_3 = L + \frac{\frac{N}{4} \times 3 - \text{pcf}}{f} \times c = 21 + \frac{148.5 - 136}{32} \times 2 = 21.78$$

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2} = (21.78 - 17.94) / 2 = 3.84 / 2 = 1.92$$

$$\text{Coefficient of Quartile Deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100 = [(21.78 - 17.94) / (21.78 + 17.94)] \times 100$$

$$= 3.84 / 39.72 \times 100 = 9.66\%$$

Mean Deviation and its Coefficient:

The mean deviation or the average deviation is defined as the mean of the absolute deviations of observations from some suitable average which may be the arithmetic mean, the median or the mode. The difference (x-average) is called deviation and when we ignore the negative sign, this deviation is written as $|X - \text{average}|$ and is read as mod deviations. The mean of these mod or absolute deviations is called the mean deviation or the mean absolute deviation. Thus for sample data in which the suitable average is the $\bar{X}$, the mean deviation (M.D) is given by the relation:

$$M.D = \frac{\sum |X - \bar{X}|}{n} \quad [\text{Mean deviation about mean}]$$

$$M.D(\text{about median}) = \frac{\sum |X - \text{Median}|}{N} \quad [\text{Mean deviation about median}]$$

$$M.D(\text{about mode}) = \frac{\sum |X - \text{Mode}|}{N} \quad [\text{Mean deviation about mode}]$$

$$M.D(\text{about constant}) = \frac{\sum |X - A|}{N} \quad [\text{Mean deviation about a constant value}]$$

$$M.D = \frac{\sum f |X - \bar{X}|}{\sum f} \quad [\text{Mean deviation about mean}]$$

$$M.D = \frac{\sum f |X - \bar{X}|}{N} \quad [\because \sum f = N]$$

$$M.D(\text{about median}) = \frac{\sum f |X - \text{Median}|}{N} \quad [\text{Mean deviation about median}]$$

$$M.D(\text{about mode}) = \frac{\sum f |X - \text{Mode}|}{N} \quad [\text{Mean deviation about mode}]$$

$$M.D(\text{about constant}) = \frac{\sum |X - A|}{N} \quad [\text{Mean deviation about a constant value}]$$

The mean deviation is a better measure of absolute dispersion than the range and the quartile deviation. A drawback in the mean deviation is that we use the absolute deviations which do not seem logical. The reason for this is that it is always equal to zero. Even if we use median or mode in place of $\bar{X}$, even then the summation will be zero or approximately zero with the result that the mean deviation

would always be either zero or close to zero. Thus the very definition of the mean deviation is possible only on the absolute deviations.

Merits and Demerits of M.D:

Merits:

1. It is simple to understand and easy to compute.
2. It is rigidly defined.
3. It is based on all items of the series.
4. It is not much affected by the fluctuations of sampling.
5. It is less affected by the extreme items.
6. It is flexible, because it can be calculated from any average.
7. It is better measure of comparison.

Demerits:

1. It is not a very accurate measure of dispersion.
2. It is not suitable for further mathematical calculation.
3. It is rarely used. It is not as popular as standard deviation.
4. Algebraic positive and negative signs are ignored. It is mathematically unsound and illogical.

Coefficient of the Mean Deviation:

A relative measure of dispersion based on the mean deviation is called the coefficient of the mean deviation or the coefficient of dispersion. It is defined as the ratio of the mean deviation to the average used in the calculation of the mean deviation. Thus

$$\text{Coefficient of M.D(about mean)} = \frac{\text{Mean Deviation from Mean}}{\text{Mean}} \times 100$$

$$\text{Coefficient of M.D(about median)} = \frac{\text{Mean Deviation from Median}}{\text{Median}} \times 100$$

$$\text{Coefficient of } M.D(\text{about Mode}) = \frac{\text{Mean Deviation from Mode}}{\text{Mode}} \times 100$$

$$\text{Coefficient of } M.D(\text{about constant}) = \frac{\text{Mean Deviation from constant value}}{\text{constant}} \times 100$$

Example 1

Find the mean deviation about the mean for the following data:

6, 7, 10, 12, 13, 4, 8, 12

Mean of the given data = $\frac{\text{Sum of all terms}}{\text{Total number of terms}}$

$$\begin{aligned}\bar{x} &= \frac{6 + 7 + 10 + 12 + 13 + 4 + 8 + 12}{8} \\ &= \frac{72}{8} \\ &= 9\end{aligned}$$

Mean deviation about mean

$$\begin{aligned}&= \frac{\sum |x_i - \bar{x}|}{8} \\ &= \frac{22}{8} \\ &= 2.75\end{aligned}$$

x_i	$x_i - \bar{x}$	$ x_i - \bar{x} $
6	$6 - 9 = -3$	$ -3 = 3$
7	$7 - 9 = -2$	$ -2 = 2$
10	$10 - 9 = 1$	$ 1 = 1$
12	$12 - 9 = 3$	$ 3 = 3$
13	$13 - 9 = 4$	$ 4 = 4$
4	$4 - 9 = -5$	$ -5 = 5$
8	$8 - 9 = -1$	$ -1 = 1$
12	$12 - 9 = 3$	$ 3 = 3$
		$\sum_{i=1}^8 x_i - \bar{x} = 22$

Coefficient of mean deviation

$$\text{Coefficient of M.D(about mean)} = \frac{\text{Mean Deviation from Mean}}{\text{Mean}} \times 100$$

$$\text{Coefficient of M.D(about mean)} = \frac{2.75}{9} \times 100 =$$

$$= 30.55\%$$

Mean deviation about Median:

Here, 4 6 7 8 10 12 12 13: Median=9

x	x-median	$x - \text{median}$
4	4-9=-5	$-5 =5$
6	6-9=-3	$-3 =3$
7	7-9=-2	$-2 =2$
8	8-9=-1	$-1 =1$
10	10-9=1	$1 =1$
12	12-9=3	$3 =3$
12	12-9=3	$3 =3$
13	13-9=4	$4 =4$
Total		$\sum x - \text{median} = 22$

$$\text{M.D(about median)} = \frac{\sum |X - \text{Median}|}{N}$$

$$= 22/8 = 2.75$$

$$\text{Coefficient of } M.D(\text{about median}) = \frac{\text{Mean Deviation from Median}}{\text{Median}} \times 100$$

$$= [2.75/8] \times 100 = 30.75\%$$

Mean deviation about Mode:

Here Mode=12

x	x-mode	$x - mode$
4	4-12=-8	$-8 =8$
6	6-12=-6	$-6 =6$
7	7-12=-5	$-5 =5$
8	8-12=-4	$-4 =4$
10	10-12=-2	$-2 =2$
12	12-12=0	$0 =0$
12	12-12=0	$0 =0$
13	13-12=1	$1 =1$
Total		$\sum x - mode = 26$

$$M.D(\text{about mode}) = \frac{\sum f|X - Mode|}{N} \quad [\text{Mean deviation about mode}]$$

$$=26/8=3.25$$

$$\text{Coefficient of } M.D(\text{about Mode}) = \frac{\text{Mean Deviation from Mode}}{\text{Mode}} \times 100$$

$$=[3.25/12] \times 100 = 27.08\%$$

Mean deviation about constant value(A):

	x-A (8)	 x - A
4	4-8=-3	 -3 =3
6	6-8=-2	 -2 =2
7	7-8=-1	 -1 =1
8	8-8=0	 0 =0
10	10-8= 2	 2 =2
12	12-8= 4	 4 =4
12	12--8= 4	 4 =4
13	13-8= 5	 5 =5
Total		$\sum x - A = 21$

$$M.D(\text{about constant}) = \frac{\sum |X - A|}{N}$$

$$=21/8=2.625$$

$$\text{Coefficient of } M.D(\text{about constant}) = \frac{\text{Mean Deviation from constant value}}{\text{constant } t} \times 100$$

$$=[2.625/8] \times 100 = 32.81\%$$

Mean deviation for group data:

$$M.D = \frac{\sum f |X - \bar{X}|}{\sum f}$$

$$M.D = \frac{\sum f |X - \bar{X}|}{N} [\because \sum f = N]$$

MEAN (OR AVERAGE) DEVIATION

- Exp: Calculate mean deviation from the FD (Grouped Data).

<i>X</i>	<i>f</i>	Class Mark (<i>x</i>)	<i>f.x</i>	<i> x - 6.57 </i>	<i>f x - 6.57 </i>
2 - 4	2	3	6	3.57	7.14
4 - 6	3	5	15	1.57	4.71
6 - 8	6	7	42	0.43	2.58
8 - 10	2	9	18	2.43	4.86
10 - 12	1	11	11	4.43	4.43
Total	$\Sigma f = 14$		$\Sigma f.x = 92$		$\Sigma f x - 6.57 = 23.72$

$$M.D = 23.72 / 14 = 1.69$$

$$\bar{x} = 92 / 14 = 6.57$$

Finding **mean deviation about Median** = $\frac{\sum f_i |x_i - M|}{\sum f_i}$

Class	Frequency	Cumulative frequency	Mid-point x_i	$ x_i - M $	$f_i x_i - M $
0 – 10	6	6	5	$ 5 - 28 = 23$	$6 \times 23 = 138$
10 – 20	7	$7 + 6 = 13$	15	$ 15 - 28 = 13$	$7 \times 13 = 91$
20 – 30	15	$13 + 15 = 28$	25	$ 25 - 28 = 3$	$15 \times 3 = 45$
30 – 40	16	$28 + 16 = 44$	35	$ 35 - 28 = 7$	$16 \times 7 = 112$
40 – 50	4	$44 + 4 = 48$	45	$ 45 - 28 = 17$	$4 \times 17 = 68$
50 – 60	2	$48 + 2 = 50$	55	$ 55 - 28 = 27$	$2 \times 27 = 54$
	$\sum f_i = 50$			$\sum f_i x_i - M = 508$	

$$\sum f_i = 50 \text{ \& \; } \sum f_i |x_i - M| = 508$$

$$\therefore \text{Mean Deviation (M)} = \frac{\sum f_i |x_i - M|}{\sum f_i}$$

$$= \frac{508}{50}$$

$$= \mathbf{10.16}$$

Example 1. Compute M.D. and Coefficient of M.D. from mean and median for following series.

3	7	12	14	15	18	22
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Solution

$$(1) \bar{X} = \frac{\sum X}{N}; \sum X = 3 + 7 + 12 + 14 + 15 + 18 + 22 = 91 \therefore \bar{X} = \frac{91}{7} = 13 \quad [N = 7]$$

X	dy = X - $\bar{X}$ ($\pm$ signs ignored)
3	10
7	6
12	1
14	1
15	2
18	5
22	9
	$\Sigma dy = 34$

$$\text{Now } \Sigma dy = 34; \quad N = 7 \text{ And M.D.} = \frac{\Sigma dy}{N}$$

$$= \frac{34}{7} = 4.86. \therefore \text{M.D. from Mean} = 4.86.$$

And Coefficient of Mean Deviation

$$= \frac{\text{M.D. from Mean}}{\text{Mean}} = \frac{4.86}{13} = .374.$$

$$(2) N = 7; \therefore \text{Median is 4th term} = 14.$$

X	($\pm$ signs ignored) dy = X - M
3	11
7	7
12	2
14	0
15	1
18	4
22	8
	$\Sigma dy = 33$

$$\text{Now } \Sigma dy = 33, \quad N = 7$$

$$\text{And M.D.} = \frac{\Sigma dy}{N} = \frac{33}{7} = 4.71$$

Coefficient of Mean Deviation from Median

$$= \frac{\text{M.D. from Median}}{\text{Median}} = \frac{4.71}{14} = .336$$

Example 2. Compute Mean Deviation from $\bar{X}$, M and Z for following data (Using direct Method)

7	9	13	13	15	17	19	21	23
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Solution

$$(1) \text{ From } \bar{X}: \bar{X} = \frac{\sum X}{N} = \frac{7 + 9 + 13 + 13 + 15 + 17 + 19 + 21 + 23}{9} = \frac{137}{9} = 15.22.$$

$$\text{And M.D.} = \frac{\sum X_1 - \sum X_2 - (\sum N_1 - \sum N_2)(\bar{X} \text{ or } M \text{ or } Z)}{N} \quad \dots(1)$$

$$\sum X_1 = 17 + 19 + 21 + 23 = 80; \quad \sum N_1 = 4$$

$$\sum X_2 = 7 + 9 + 13 + 13 + 15 = 57; \quad \sum N_2 = 5; \quad N = 9$$

$$\therefore \text{ From (1) M.D.} = \frac{80 - 57 - (4 - 5) 15.22}{9} = \frac{23 + 15.22}{9} = \frac{38.22}{9} = 4.25.$$

$$\text{Coefficient of M.D.} = \frac{4.25}{15.22} = .279$$

$$\left[\frac{\text{M.D.}}{\bar{X}} \right]$$

Example : 3.

Class Intervals of scores	Mid point X	f	fX	x $(X-M)$	fx
80-84	82	1	82	25	25
75-79	77	3	231	20	60
70-74	72	5	360	15	75
65-69	67	7	469	10	70
60-64	62	8	496	5	40
55-59	57	11	627	0	0
50-54	52	9	468	-5	-45
45-49	47	7	329	-10	-70
40-44	42	6	252	-15	-90
35-39	37	2	74	-20	-40
30-34	32	1	32	-25	-25
		$N=60$	3420		$\Sigma fx = 540$

$$\text{Mean} = \frac{\Sigma fX}{N} = \frac{3420}{60} = 57$$

$$AD \text{ or } MD = \frac{\Sigma |fx|}{N} = \frac{540}{60} = 9$$

Standard Deviation : Standard Deviation shows the variation in data. If the data is close together, the standard deviation will be small. If the data is spread out, the standard deviation will be large. Standard Deviation is often denoted by the lowercase Greek letter sigma, σ . Let $x_1, x_2, \dots, x_n$ observations and their mean $\bar{x}$ then

deviations $x_1 - \bar{x}, x_2 - \bar{x}, \dots, x_n - \bar{x}$

square of deviations, $(x_1 - \bar{x})^2, (x_2 - \bar{x})^2, \dots, (x_n - \bar{x})^2$

sum of square of deviation $(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2$

$$= \sum (x_i - \bar{x})^2$$

Standard deviation
$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Steps to Finding Standard Deviation

- 1) Find the mean of the data. 2) Subtract the mean from each value. 3) Square each deviation of the mean. 4) Find the sum of the squares. 5) Divide the total by the number of items. 6) Take the square root.

So simply standard deviation is called Root –Mean-Square -Deviation

Variance

The mean of the squares of the deviations from the mean is called **Variance**.

It is denoted by σ^2 (read as sigma square).

Variance = Mean of squares of deviations

$$\begin{aligned}\text{Variance} &= \text{Mean of squares of deviations} \\ &= \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n} \\ \text{Variance } \sigma^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}\end{aligned}$$

Example : The number of televisions sold in each day of a week are 7,9,12,15,5 4, 11. Find its standard deviation.

We know,

$$\text{Standard deviation } \sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\begin{aligned}\bar{x} &= \frac{7+9+12+15+5+4+11}{7} \\ &= \frac{63}{7} = 9\end{aligned}$$

X_i	Deviation= $(x_i - \bar{x})$	$(x_i - \bar{x})^2$
7	7-9=-2	$(-2)^2=4$
9	9-9=0	0
12	12-9=3	$(3)^2= 9$
15	15-9=6	$(6)^2= 36$
5	5-9=-4	$(-4)^2=16$
4	4-9=-5	$(-5)^2=25$
11	11-9=2	$(2)^2= 4$
Total		$\sum (x_i - \bar{x})^2$ =94

Standard deviation $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$

$$= \sqrt{94/7} = \sqrt{13.428} = 3.664$$

Variance = $\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = 13.42$

Mean = 9 **sd** = 3.664 **95% CI** = 5.34 to 12.66

Or, $\bar{x} \pm \sigma$

Mean = 9

S. D = $\sigma = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$

$$\begin{aligned} (S.D)^2 = \sigma^2 &= \frac{1}{n} \sum (x_i - \bar{x})^2 \\ &= \frac{1}{7} [(7 - 9)^2 + (9 - 9)^2 + (12 - 9)^2 + (15 - 9)^2 + (5 - 9)^2 + (4 - 9)^2 + (11 - 9)^2] \\ &= \frac{1}{7} (4 + 0 + 9 + 36 + 16 + 25 + 4) = \frac{1}{7} \times 94 = 13.428 \end{aligned}$$

S.D = $\sqrt{13.428} = 3.664$

Another methods(Direct method)

Example : The number of televisions sold in each day of a week are 7,9,12,15,5 4, 11. Find its standard deviation.

We know,

$$S.D = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$$

x_i	x_i^2
7	49
9	81
12	144
15	225
5	25
4	16
11	121
$\sum x_i = 63$	$\sum x_i^2 = 661$

$$S.D = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$$

$$= \sqrt{\frac{661}{7} - (63/7)^2} = \sqrt{(94.42 - 81)} = \sqrt{13.42} = 3.66$$

Example : The amount of rainfall in a particular season for 6 days are given as 17.8 cm, 19.2 cm, 16.3 cm, 12.5 cm, 12.8 cm and 11.4 cm. Find its standard deviation.

Solution Arranging the numbers in ascending order we get, 11.4, 12.5, 12.8, 16.3, 17.8, 19.2.

Number of observations $n = 6$

$$\text{Mean} = \frac{11.4 + 12.5 + 12.8 + 16.3 + 17.8 + 19.2}{6} = \frac{90}{6} = 15$$

x_i	$d_i = x_i - \bar{x}$ $= x - 15$	d_i^2
11.4	-3.6	12.96
12.5	-2.5	6.25
12.8	-2.2	4.84
16.3	1.3	1.69
17.8	2.8	7.84
19.2	4.2	17.64
		$\Sigma d_i^2 = 51.22$

$$\begin{aligned} \text{Standard deviation } \sigma &= \sqrt{\frac{\Sigma d_i^2}{n}} \\ &= \sqrt{\frac{51.22}{6}} = \sqrt{8.53} \end{aligned}$$

Hence, $\sigma \simeq 2.9$

$$\text{Variance} = \sigma^2 = 8.53$$

Another method or Direct Method

$$\begin{aligned}
 \text{Standard deviation } \sigma &= \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} \\
 &= \sqrt{\frac{\sum (x_i^2 - 2x_i\bar{x} + \bar{x}^2)}{n}} \\
 &= \sqrt{\frac{\sum x_i^2}{n} - 2\bar{x} \frac{\sum x_i}{n} + \frac{\bar{x}^2}{n} \times (1 + 1 + \dots \text{to } n \text{ times})} \\
 &= \sqrt{\frac{\sum x_i^2}{n} - 2\bar{x} \times \bar{x} + \frac{\bar{x}^2}{n} \times n} = \sqrt{\frac{\sum x_i^2}{n} - 2\bar{x}^2 + \bar{x}^2} = \sqrt{\frac{\sum x_i^2}{n} - \bar{x}^2} \\
 \text{Standard deviation, } \sigma &= \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}
 \end{aligned}$$

Note

- While computing standard deviation, arranging data in ascending order is not mandatory.
- If the data values are given directly then to find standard deviation we can use

$$\text{the formula } \sigma = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$$

- If the data values are not given directly but the squares of the deviations from the mean of each observation is given then to find standard deviation we can use

$$\text{the formula } \sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

Example : The number of televisions sold in each day of a week are 13, 8, 4, 9, 7, 12, 10.

Find its standard deviation.

Solution

x_i	x_i^2
13	169
8	64
4	16
9	81
7	49
12	144
10	100
$\Sigma x_i = 63$	$\Sigma x_i^2 = 623$

Standard deviation

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2} \\ &= \sqrt{\frac{623}{7} - \left(\frac{63}{7}\right)^2} \\ &= \sqrt{89 - 81} = \sqrt{8} \\ \text{Hence, } \sigma &\simeq 2.83\end{aligned}$$

Variance = (sd)² = 8

Example:

The scores of a cricket player for 10 innings are given below. .

Cricketer	101	40	0	13	67	78	6	90	0	3
-----------	-----	----	---	----	----	----	---	----	---	---

Calculation of S D .

Cricketer A	
X	$(X - \bar{X})^2$
101	3745.44
40	0.04
0	1584.04
13	718.24
67	739.84
78	1459.24
6	1142.44
90	2520.04
0	1584.04
3	1354.24
$\Sigma X = 398$	$\Sigma(X - \bar{X})^2 = 14847.6$

Here $\bar{X} = \frac{\Sigma X}{n} = \frac{398}{10} = 39.8$

$$S.D = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n}} = \sqrt{\frac{14847.6}{10}} = 38.53$$

or

Example

The demand of a commodity in a town is given below:

Town A	36	42	18	35	48	62
--------	----	----	----	----	----	----

Solution:

Calculation of S .D.

Town A	
X	$(X - \bar{X})^2$
36	17.39
42	3.35
18	491.51
35	26.73
48	61.31
62	476.55
$\Sigma X = 241$	$\Sigma(X - \bar{X})^2 = 1076.84$

Here $\bar{X} = \frac{\Sigma X}{n} = \frac{241}{6} = 40.17$

$$\text{S.D. (Town A)} = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n}} = \sqrt{\frac{1076.84}{6}} = 13.4$$

$$\text{Variance} = (SD)^2 = 179.56$$

Calculation of Standard deviation for grouped data

$$\text{Standard deviation} = \sigma = \sqrt{\frac{\Sigma fi(Xi - \bar{X})^2}{N}}$$

where, $N = \sum f_i$

$$\text{Variance} = \sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$$

OR,

$$\text{Standard deviation} = \sigma = \sqrt{\frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N} \right)^2}$$

$$\text{Variance} = (SD)^2 = \sigma^2$$

Example Calculate the S.D and variance from the data given below:

Daily wages	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of workers	2	7	10	5	3

Solution:

Calculation of S.D and Variance

Daily wages	X_i	No. of workers (f_i)	$(X - \bar{X})$	$f_i(X - \bar{X})^2$
0 – 10	5	2	5-25=-20	800
10 – 20	15	7	-10	700
20 – 30	25	10	0	0
30 – 40	35	5	10	500
40 – 50	45	3	20	1200
		$N = 27$		$\sum f_i(X - \bar{X})^2 = 3200$

$$\text{Mean} = (\bar{X}) = \frac{\sum f_i X_i}{N} = \frac{675}{27} = 25 \quad [\text{Ans}]$$

$$\text{Standard deviation} = \sigma = \sqrt{\frac{\sum f_i (X_i - \bar{X})^2}{N}}$$

$$\sigma = \sqrt{\frac{3200}{27}}$$

$$= 10.886$$

$$\text{Variance} = (SD)^2 = \sigma^2 = 118.52$$

OR,

Calculate the S.D and variance from the data given below:

Daily wages	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of workers	2	7	10	5	3

Solution:

$$\text{Standard deviation} = \sigma = \sqrt{\frac{\sum f_i X_i^2}{N} - \left(\frac{\sum f_i X_i}{N} \right)^2}$$

$$\text{Variance} = (SD)^2 = \sigma^2$$

Calculation of S.D and Variance

Daily wages	X_i	No.of workers(f_i)	$f_i X_i$	$f_i X_i^2$
0 – 10	5	2	10	250
10 – 20	15	7	105	1575
20 – 30	25	10	250	6250
30 – 40	35	5	175	6125
40 – 50	45	3	135	6075
		$N = 27$	$\Sigma f_i X_i = 675$	$\Sigma f_i X_i^2 = 20075$

$$\text{Standard deviation} = \sigma = \sqrt{\frac{20075}{27} - \left(\frac{675}{27}\right)^2} = \sqrt{118.52} = 10.886.$$

$$\text{Variance} = (SD)^2 = \sigma^2 = 118.52$$

Example:

A sample of daily production was organized into the following distribution:

Daily Production	Frequency
80 – 90	5
90 – 100	9
100 – 110	20
110 – 120	8
120 – 130	6
130 – 140	2

Calculate the standard deviation and variance.

Solution:

Calculation of S.D. and Variance

Marks	fi	Xi	fiXi	fiXi ²
80 – 90	5	85	425	36125
90 – 100	9	95	855	81225
100 – 110	20	105	2100	220500
110 – 120	8	115	920	105800
120 – 130	6	125	750	93750
130 – 140	2	135	270	36450
	N = 50		ΣfX = 5320	ΣfX ² = 573850

$$) \text{ S.D.} = \sqrt{\frac{\Sigma fX^2}{N} - \left(\frac{\Sigma fX}{N}\right)^2} = \sqrt{\frac{573850}{50} - \left(\frac{5320}{50}\right)^2} = 12.49 \text{ [Ans.]}$$

$$\text{Variance} = (\text{S.D.})^2 = (12.49)^2 = 156.04 \text{ [Ans.]}$$

Example

48 students were asked to write the total number of hours per week they spent on watching television. With this information find the standard deviation of hours spent for watching television.

<i>x</i>	6	7	8	9	10	11	12
<i>f</i>	3	6	9	13	8	5	4

Solution

<i>xi</i>	<i>fi</i>	<i>fixi</i>	<i>fi xi²</i>
<i>6</i>	<i>3</i>	<i>18</i>	<i>108</i>
<i>7</i>	<i>6</i>	<i>48</i>	<i>294</i>
<i>8</i>	<i>9</i>	<i>72</i>	<i>576</i>
<i>9</i>	<i>13</i>	<i>112</i>	<i>1053</i>
<i>10</i>	<i>8</i>	<i>80</i>	<i>800</i>
<i>11</i>	<i>5</i>	<i>55</i>	<i>605</i>

12	4	48	576
Total		$\sum \text{fixi} = 432$	$\sum \text{fixi}^2 = 4012$

$$\text{Standard deviation} = \sigma = \sqrt{\frac{\sum \text{fixi}^2}{N} - \left(\frac{\sum \text{fixi}}{N}\right)^2}$$

$$\text{Standard deviation} = \sigma = \sqrt{\frac{4012}{48} - \left(\frac{432}{48}\right)^2}$$

$$= \sqrt{83.583 - 81} = \sqrt{2.583} = 1.607$$

$$\text{Variance} = (SD)^2 = \sigma^2 = 2.583$$

4.16.1 Merits and Demerits of Standard Deviation:

Merits:

1. It is rigidly defined and its value is always definite and based on all the observations and the actual signs of deviations are used.
2. As it is based on arithmetic mean, it has all the merits of arithmetic mean.
3. It is the most important and widely used measure of dispersion.
4. It is possible for further algebraic treatment.
5. It is less affected by the fluctuations of sampling and hence stable.
6. It is the basis for measuring the coefficient of correlation and sampling.

Demerits:

1. It is not easy to understand and it is difficult to calculate.
2. It gives more weight to extreme values because the values are squared up.
3. As it is an absolute measure of variability, it cannot be used for the purpose of comparison.

Combined Mean and Standard deviation:

Example

For two factories A and B belonging to the same industry, following details are available:

	Factory A	Factory B
No. of wages earners	100(N ₁)	80(nN ₂)
Average wage monthly	Tk 4600	Tk 4900
S.D.	Tk 500	Tk 400

What is the mean and standard deviation of all the workers in two factories taken together?

Solution:

Combined Mean:

$$\begin{aligned}\bar{X}_{12} &= \frac{N_1\bar{X}_1 + N_2\bar{X}_2}{N_1 + N_2} \\ \Rightarrow \bar{X}_{12} &= \frac{(4600 \times 100) + (4900 \times 80)}{100 + 80} \\ \Rightarrow \bar{X}_{12} &= 4733.33(\text{Ans.})\end{aligned}$$

Combined standard deviation (S.D.):

$$\begin{aligned}\sigma_{12} &= \sqrt{\frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1d_1^2 + N_2d_2^2}{N_1 + N_2}} \\ &= \sqrt{\frac{100(250000) + 80(160000) + 100(17777) + 80(27779)}{100 + 80}} \\ &= 481.89 \text{ (Ans.)}\end{aligned}$$

Where,

$$\begin{aligned}d_1^2 &= |\bar{X}_1 - \bar{X}_{12}|^2 \\ &= |4600 - 4733.33|^2 \\ &= 17777\end{aligned}$$

$$\begin{aligned}d_2^2 &= |\bar{X}_2 - \bar{X}_{12}|^2 \\ &= |4900 - 4733.33|^2 \\ &= 27779\end{aligned}$$

$$\sigma_1^2 = (500)^2 = 250000$$

$$\sigma_2^2 = (400)^2 = 160000$$

Example

For a group of 50 male workers, the mean and standard deviation of their wages are Tk 63 and Tk 9 respectively. For a group of 40 female workers, these are Tk 54 and Tk 6 respectively. Find the mean and standard deviation of daily wages for the combined group of 90 workers.

Solution: Given

$$N_1 = 50$$

$$N_2 = 40$$

$$\bar{X}_1 = 63$$

$$\bar{X}_2 = 54$$

$$\sigma_1 = 9$$

$$\sigma_2 = 6$$

$$(i) \text{ Combined Mean } (\bar{X}_{12}) = \frac{(N_1 \times \bar{X}_1) + (N_2 \times \bar{X}_2)}{N_1 + N_2}$$

$$= \frac{(50 \times 63) + (40 \times 54)}{50 + 40}$$

$$= 59 \text{ [Ans.]}$$

$$\begin{aligned}
 \text{Combined S.D. } (\sigma_{12}) &= \sqrt{\frac{N_1\sigma_1^2 + N_2\sigma_2^2 + N_1d_1^2 + N_2d_2^2}{N_1 + N_2}} \\
 &= \sqrt{\frac{50(81) + 40(36) + 50(16) + 40(25)}{50 + 40}} \\
 &= 9 \text{ (Ans.)}
 \end{aligned}$$

Where,

$$d_1^2 = |\bar{X}_{12} - \bar{X}_1|^2$$

$$= |59 - 63|^2$$

$$= 16$$

$$d_2^2 = |\bar{X}_{12} - \bar{X}_2|^2$$

$$= |59 - 54|^2$$

$$= 25$$

$$\sigma_1^2 = (9)^2 = 81$$

$$\sigma_2^2 = (6)^2 = 36$$

Example Find the standard deviation of the following data 7, 4, 8, 10, 11. Add 3 to all the values then find the standard deviation for the new values.

Solution Arranging the values in ascending order we get, 4, 7, 8, 10, 11 and $n = 5$

x_i	x_i^2
4	16
7	49
8	64
10	100
11	121
$\Sigma x_i = 40$	$\Sigma x_i^2 = 350$

Standard deviation

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2} \\ &= \sqrt{\frac{350}{5} - \left(\frac{40}{5}\right)^2} \\ \sigma &= \sqrt{6} \simeq 2.45\end{aligned}$$

When we add 3 to all the values, we get the new values as 7, 10, 11, 13, 14.

x_i	x_i^2
7	49
10	100
11	121
13	169
14	196
$\Sigma x_i = 55$	$\Sigma x_i^2 = 635$

Standard deviation

$$\begin{aligned}\sigma &= \sqrt{\frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2} \\ &= \sqrt{\frac{635}{5} - \left(\frac{55}{5}\right)^2} \\ \sigma &= \sqrt{6} \simeq 2.45\end{aligned}$$

From the above, we see that the standard deviation will not change when we add some fixed constant to all the values.

Example Find the standard deviation of the data 2, 3, 5, 7, 8. Multiply each data by 4. Find the standard deviation of the new values.

Solution Given, $n = 5$

x_i	x_i^2
2	49
3	9
5	25
7	49
8	64
$\Sigma x_i = 25$	$\Sigma x_i^2 = 151$

Standard deviation

$$\sigma = \sqrt{\frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2}$$

$$\sigma = \sqrt{\frac{151}{5} - \left(\frac{25}{5}\right)^2} = \sqrt{30.2 - 25} = \sqrt{5.2} \simeq 2.28$$

When we multiply each data by 4, we get the new values as 8, 12, 20, 28, 32.

x_i	x_i^2
8	64
12	144
20	400
28	784
32	1024
$\Sigma x_i = 100$	$\Sigma x_i^2 = 2416$

Standard deviation

$$\sigma = \sqrt{\frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2}$$

$$= \sqrt{\frac{2416}{5} - \left(\frac{100}{5}\right)^2} = \sqrt{483.2 - 400} = \sqrt{83.2}$$

$$\sigma = \sqrt{16 \times 5.2} = 4\sqrt{5.2} \simeq 9.12$$

From the above, we see that when we multiply each data by 4 the standard deviation also get multiplied by 4.

Example Find the mean and variance of the first n natural numbers.

Solution

$$\begin{aligned}
 \text{Mean } \bar{x} &= \frac{\text{Sum of all the observations}}{\text{Number of observations}} \\
 &= \frac{\sum x_i}{n} = \frac{1 + 2 + 3 + \dots + n}{n} = \frac{n(n+1)}{2 \times n} \\
 \text{Mean } \bar{x} &= \frac{n+1}{2} \\
 \text{Variance } \sigma^2 &= \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 \left[\begin{aligned} \sum x_i^2 &= 1^2 + 2^2 + 3^2 + \dots + n^2 \\ (\sum x_i)^2 &= (1 + 2 + 3 + \dots + n)^2 \end{aligned} \right] \\
 &= \frac{n(n+1)(2n+1)}{6 \times n} - \left[\frac{n(n+1)}{2 \times n} \right]^2 \\
 &= \frac{2n^2 + 3n + 1}{6} - \frac{n^2 + 2n + 1}{4} \\
 \text{Variance } \sigma^2 &= \frac{4n^2 + 6n + 2 - 3n^2 - 6n - 3}{12} = \frac{n^2 - 1}{12}.
 \end{aligned}$$

Example

The mean and standard deviation of 15 observations are found to be 10 and 5 respectively. On rechecking it was found that one of the observation with value 8 was incorrect. Calculate the correct mean and standard deviation if the correct observation value was 23?

Solution

$$n = 15, \bar{x} = 10, \sigma = 5; \quad \bar{x} = \frac{\sum x}{n};$$

$$\sum x = 15 \times 10 = 150$$

Wrong observation value = 8, Correct observation value = 23.

Correct total = $150 - 8 + 23 = 165$

$$\text{Correct mean } \bar{x} = \frac{165}{15} = 11$$

$$\text{Standard deviation } \sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$\text{Incorrect value of } \sigma = 5 = \sqrt{\frac{\Sigma x^2}{15} - (10)^2}$$

$$25 = \frac{\Sigma x^2}{15} - 100 \text{ gives, } \frac{\Sigma x^2}{15} = 125$$

$$\text{Incorrect value of } \Sigma x^2 = 1875$$

$$\text{Correct value of } \Sigma x^2 = 1875 - 8^2 + 23^2 = 2340$$

$$\text{Correct standard deviation } \sigma = \sqrt{\frac{2340}{15} - (11)^2}$$

$$\sigma = \sqrt{156 - 121} = \sqrt{35} \quad \sigma \simeq 5.9$$

Coefficient of Variation: In [statistics](#), the **coefficient of variation (CV)**, also known as **relative standard deviation (RSD)**, is a [standardized](#) measure of [dispersion](#) of a [probability distribution](#) or [frequency distribution](#). It is often expressed as a percentage, and is defined as the ratio of the [standard deviation](#) σ to the [mean](#) $\bar{X}$. The most important of all the relative measure of dispersion is the coefficient of variation. This word is variation not variance. There is no such thing as coefficient of variance. The coefficient of variation (C.V) is defined as:

$$\text{Coefficient of variation (C.V)} = \frac{\text{S.D.}}{\bar{X}} \times 100 =$$

(standard deviation / mean) $\times 100$

Example.1. The standard deviation and mean of a data are 6.5 and 12.5 respectively. Find the coefficient of variation.

Solution :

S.D $\sigma = 6.5$ mean $\bar{x} = 12.5$

$$\begin{aligned} \text{Coefficient of variation C.V} &= \frac{\sigma}{\bar{x}} \times 100 \\ &= \frac{6.5}{12.5} \times 100 \end{aligned}$$

Ans : Coefficient of variation = 52 %

Example:2. Find the coefficient of variation of 24, 26, 33, 37, 29, 31.

Solution :

Arrange in ascending order 24, 26, 29, 31, 33, 37.

$$\bar{x} = \frac{24 + 26 + 29 + 31 + 33 + 37}{6} = \frac{180}{6} = 30$$

$$\bar{x} = 30$$

x_i	$d_i = x_i - \bar{x}$ $= x_i - 30$	d_i^2
24	-6	36
26	-4	16
29	-1	1
31	1	1
33	3	9
37	7	49
$\Sigma d_i = 0$		$\Sigma d_i^2 = 112$

$$\sigma = \frac{\Sigma d_i^2}{n}$$

$$= \sqrt{\frac{112}{6}} = \sqrt{18.67}$$

$$= 4.32$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{4.32}{30} \times 100 = 14.4$$

Answer :

Coefficient of variation = 14.4 %

Example 3: The time taken (in minutes) to complete a homework by 8 students in a day are given by 38, 40, 47, 44, 46, 43, 49, 53. Find the coefficient of variation.

Solution :

Let us arrange in ascending order 38, 40, 43, 44, 46, 47, 49, 53

$$\bar{x} = \frac{38 + 40 + 43 + 44 + 46 + 47 + 49 + 53}{8} = \frac{360}{8} = 45$$

$$\bar{x} = 45$$

x_i	$d_i = x_i - \bar{x}$ $= x_i - 45$	d_i^2
38	-7	49
40	-5	25
43	-2	4
44	-1	1
46	1	1
47	2	4
49	4	16
53	8	64
$\Sigma d_i = 0$		$\Sigma d_i^2 = 164$

$$\sigma = \sqrt{\frac{\Sigma d_i^2}{n}}$$
$$= \sqrt{\frac{164}{8}} = \sqrt{20.5}$$

$$\sigma = 4.53$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100 = \frac{4.53}{45} \times 100$$

$$= 0.1007 \times 100 = 10.07 \%$$

Answer : Coefficient of variation = 10.07 %

Example3: The total marks scored by two students Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance?

Solution :

$$n = 5$$

Vidhya	Sathya
total marks $\sum x = 480$	total marks $\sum x = 460$
S.D $\sigma = 2.4$	S.D $\sigma = 4.6$
$\bar{x} = \frac{\sum x}{n} = \frac{480}{5}$	$\bar{x} = \frac{\sum x}{n} = \frac{460}{5}$
$\bar{x} = 96$	$\bar{x} = 92$
$C.V = \frac{\sigma}{\bar{x}} \times 100$	$C.V = \frac{\sigma}{\bar{x}} \times 100$
$= \frac{2.4}{96} \times 100 = 2.5$	$= \frac{4.6}{92} \times 100 = 5$

C.V of Vidhya < C.V of Sathya.

Answer : Vidhya is more consistent.

Example:4. The mean and standard deviation of marks obtained by 40 students of a class in three subjects Mathematics, Science and Social Science are given below.

Subject	Mean	SD
Mathematics	56	12
Science	65	14
Social Science	60	10

Which of the three subjects shows highest variation and which shows lowest variation in marks?

Solution :

Mathematics	Science	Social Science
$\bar{x} = 56 \quad \sigma = 12$ $C.V = \frac{\sigma}{\bar{x}} \times 100$ $= \frac{12}{56} \times 100$ $= 0.2143 \times 100$ $= 21.43 \%$	$\bar{x} = 65 \quad \sigma = 14$ $C.V = \frac{\sigma}{\bar{x}} \times 100$ $= \frac{14}{65} \times 100$ $= 0.2154 \times 100$ $= 21.54 \%$	$\bar{x} = 60 \quad \sigma = 10$ $C.V = \frac{\sigma}{\bar{x}} \times 100$ $= \frac{10}{60} \times 100$ $= 0.1667 \times 100$ $= 16.67 \%$

C.V Maths = 21.43 % C.V of Science = 21.54 % C.V of Social = 16.67 %

Answer : Highest variation is Science.

Lowest variation is Social Science.

Example 5: The temperature of two cities A and B in a winter season are given below.

Temperature of city A (in degree Celsius)	18	20	22	24	26
Temperature of city B (in degree Celsius)	11	14	15	17	18

Find which city is more consistent in temperature changes?

Solution :

Find the coefficient of variation for city A and city B.

City A

$$\bar{x} = \frac{18+20+22+24+26}{5}$$

$$= \frac{110}{5} = 22$$

$$\bar{x} = 22$$

x_i	$d_i = x_i - \bar{x} = x_i - 22$	d_i^2
18	-4	16
20	-2	4
22	0	0
24	2	4
26	4	16
$\Sigma d_i = 0$		$\Sigma d_i^2 = 40$

$$\sigma = \sqrt{\frac{\Sigma d_i^2}{n}}$$
$$= \sqrt{\frac{40}{5}} = \sqrt{8}$$

$$\sigma = 2.828$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{2.828}{22} \times 100$$

$$= 12.85 \%$$

C.V of city A < C.V of city B.

Answer : City A is more consistent.

City B

$$\bar{x} = \frac{11+14+15+17+18}{5}$$

$$= \frac{75}{5} = 15$$

$$\bar{x} = 15$$

x_i	$d_i = x_i - \bar{x} = x_i - 15$	d_i^2
11	-4	16
14	-1	1
15	0	0
17	2	4
18	3	9
$\Sigma d_i = 0$		$\Sigma d_i^2 = 40$

$$\sigma = \sqrt{\frac{\Sigma d_i^2}{n}}$$
$$= \sqrt{\frac{30}{5}} = \sqrt{6}$$

$$\sigma = 2.449$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{2.449}{15} \times 100$$

$$= 16.33 \%$$

Example:6 For two firms A and B, belonging to the same industry, gives the following results:

	Firm A	Firm B
No. of employees	100	200
Average wage monthly	Tk 2400	Tk 1800
S.D.	Tk 60	Tk 80

- (i) Which firm pays out larger amount as wages?
- (ii) Which firm shows greater variability in the distribution of wages?

Solution:

(i) Total wages (Firm A) = $(2400 \times 100) = 240000$

Total wages (Firm B) = $(1800 \times 200) = 360000$

So the firm B pays out large amount as wages.

(ii) **Calculation of coefficient of variation:**

$$\text{Firm A: C.V.} = \frac{\sigma}{\bar{X}} \times 100 = \frac{60}{2400} \times 100 = 2.5\%$$

$$\text{Firm B: C.V.} = \frac{\sigma}{\bar{X}} \times 100 = \frac{80}{1800} \times 100 = 4.44\%$$

Since $CV(\text{firm B}) > CV(\text{firm A})$, so firm B has greater variation in the distribution of wages per employee.

Example 7

For two firms A and B belonging to same industry, the following details are available:

	Firm A	Firm B
Number of employees	100	200
Average wages per week	Tk 240	Tk 170
Standard deviation of the wages per week	Tk 6	Tk 8

(i) Find which firm pays out larger amount as weekly wages.

(ii) Which firm shows greater variability in the distribution of wages and which firm is better and why?

Solution:

(i) Total payment (Firm A) = $(240 \times 100) = 24000$

Total payment (Firm B) = $(170 \times 200) = 34000$

So the firm B pays out large amount as weekly wages.

$$(ii) \quad CV (\text{Firm A}) = \frac{\sigma}{\bar{X}} \times 10 = \frac{6}{240} \times 100 = 2.5\%$$

$$CV (\text{Firm B}) = \frac{\sigma}{\bar{X}} \times 10 = \frac{8}{170} \times 100 = 4.71\%$$

Since $CV (\text{firm B}) > CV (\text{firm A})$, so firm B has greater variation in the distribution of wages per employee. But Firm A is better, because it has more uniformity in the distribution of wages.

Example: 8

A potential investor wants to purchase shares of stocks in one of the two companies A and B which show greater variability in price. Following is the sample prices (in Taka) of the share of the companies:

Share price of Company A	315	320	350	340	360	365	355	370	372	378	410	390
Share price of Company B	1542	1522	1534	1532	1545	1530	1536	1530	1535	1550	1544	1538

Which share does the investor buy and why?

Solution:

Calculation of Mean and Coefficient of Variation

Company A		Company B	
X	$(X - \bar{x})^2$	X	$(X - \bar{x})^2$
315	2025	1542	36
320	1600	1522	196

350	100	1534	4
340	400	1532	16
360	0	1545	81
365	25	1530	36
355	25	1536	0
370	100	1530	36
372	144	1535	1
378	324	1550	196
410	2500	1544	64
390	900	1538	4
$\Sigma X=4325$	$\Sigma(X-\bar{x})^2=8143$	$\Sigma X=18438$	$\Sigma(X-\bar{x})^2=670$

Company A:

—

$$\bar{x} = \frac{\Sigma X}{N} = \frac{4325}{12} = 360$$

$$\sigma = \sqrt{\frac{\Sigma(X-\bar{X})^2}{N}} = \sqrt{\frac{8143}{12}} = 26.05$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{26.05}{360} \times 100 = 7.24\%$$

Company B:

$$\bar{x} = \frac{\Sigma X}{N} = \frac{18438}{12} = 1536$$

$$\sigma = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}} = \sqrt{\frac{670}{12}} = 7.47$$

$$C.V = \frac{\sigma}{\bar{X}} \times 100 = \frac{7.47}{1536} \times 100 = 0.49\%$$

Comment: Here the company A shows the greater variability than company B. Since company A has a greater variability upon its share price, so it is wise to buy the shares of company B.

Example:9

The demand of a commodity in two different towns are given below:

Town A	36	42	18	35	48	62
Town B	38	28	25	42	37	40

In which town, the demand condition is more stable and why?

Solution:

Calculation of C.V.

Town A		Town B	
X	$(X - \bar{X})^2$	Y	$(Y - \bar{Y})^2$
36	17.39	38	9

42	3.35	28	49
18	491.51	25	100
35	26.73	42	49
48	61.31	37	4
62	476.55	40	25
$\Sigma X = 241$	$\Sigma(X - \bar{X})^2 = 1076.84$	$\Sigma Y = 210$	$\Sigma(Y - \bar{Y})^2 = 236$

$$\text{Here } \bar{X} = \frac{\Sigma X}{n} = \frac{241}{6} = 40.17$$

$$\bar{Y} = \frac{\Sigma Y}{n} = \frac{210}{6} = 35$$

$$\text{S.D. (Town A)} = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n}} = \sqrt{\frac{1076.84}{6}} = 13.4$$

$$\text{S.D. (Town B)} = \sqrt{\frac{\Sigma(Y - \bar{Y})^2}{n}} = \sqrt{\frac{236}{6}} = 6.27$$

$$\text{C.V. (Town A)} = \frac{\text{S.D.}}{\text{Mean}} \times 100 = \frac{13.4}{40.17} \times 100 = 33.36\%$$

$$\text{C.V. (Town B)} = \frac{\text{S.D.}}{\text{Mean}} \times 100 = \frac{6.27}{35} \times 100 = 17.91\%$$

Comment: Since C.V. (Town B) < C.V. (Town A), so Town B is more stable.

Example: 10

Lives of two model of refrigerators in a recent survey are:

Life (No. of years)	Model A	Model B
0 – 2	5	2
2 – 4	16	7
4 – 6	13	12
6 – 8	7	19
8 – 10	5	9
10 – 12	4	1

What is the average life of each model of these refrigerators? Which model has greater uniformity?

Solution:

Calculation of Mean and coefficient of Variation

Life	m.p.	(X-5)/2	Model A			Model B		
	X	d	F	fd	fd ²	f	fd	fd ²
0-2	1	-2	5	-10	20	2	-4	8
2-4	3	-1	16	-16	16	7	-7	7
4-6	5	0	13	0	0	12	0	0
6-8	7	1	7	7	7	19	19	19
8-10	9	2	5	10	20	9	18	36
10-12	11	3	4	12	36	1	3	9
			N =	Σfd =	Σfd ² =	N =		
			50	3	99	50		

Model

A:

$$\bar{x} = A + \frac{\Sigma fd}{N} \times i = 5 + \frac{3}{50} \times 2 = 5.12$$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times i = \sqrt{\frac{99}{50} - \left(\frac{3}{50}\right)^2} \times 2 = 2.81$$

$$C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{2.81}{5.12} \times 100 = 54.9\%$$

Model

B:

$$\bar{x} = A + \frac{\Sigma fd}{N} \times i = 5 + \frac{29}{50} \times 2 = 6.16$$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times i = \sqrt{\frac{99}{50} - \left(\frac{29}{50}\right)^2} \times 2 = 2.23$$

$$C.V. = \frac{\sigma}{\bar{x}} \times 100 = \frac{2.23}{6.16} \times 100 = 36.2\% \text{ [Ans.]}$$

(i) The average life of model A = 5.12 years

The average life of model A = 6.16 years

(ii) Since CV (model B) < CV (model A), so model B has greater uniformity.

2. The standard deviation and coefficient of variation of a data are 1.2 and 25.6 respectively. Find the value of mean.

Solution :

$$\text{S.D } \sigma = 1.2 \quad \text{C.V} = 25.6$$

$$\text{C.V} = \frac{\sigma}{\bar{x}} \times 100$$

$$25.6 = \frac{1.2}{\bar{x}} \times 100$$

$$\bar{x} = \frac{1.2}{25.6} \times 100$$

$$= 4.69$$

Answer : mean = 4.69

3. If the mean and coefficient of variation of a data are 15 and 48 respectively, then find the value of standard deviation.

Solution :

$$\bar{x} = 15 \quad \text{C.V} = 48$$

$$\text{C.V} = \frac{\sigma}{\bar{x}} \times 100$$

$$48 = \frac{\sigma}{15} \times 100$$

$$\sigma = \frac{48 \times 15}{100} = 7.2$$

Answer : Standard deviation = 7.2

4. If $n = 5$, $\bar{x} = 6$, $\Sigma x^2 = 765$, then calculate the coefficient of variation.

Solution :

$$n = 5 \quad \bar{x} = 6$$

$$\Sigma x^2 = 765$$

$$\begin{aligned} \sigma &= \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \sqrt{\frac{\Sigma x^2}{n} - (\bar{x})^2} \\ &= \sqrt{\frac{765}{5} - 6^2} = \sqrt{153 - 36} = \sqrt{117} \end{aligned}$$

$$\sigma = 10.817$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{10.817}{6} \times 100$$

$$= 1.8028 \times 100 = 180.28\%$$

Answer :

Coefficient of variation = 180.28 %